BSDS V: Regression Techniques

Problems Set 1

1 Random Vectors

1. Show the following equalities:
   (a) $\text{Var}(\underset{\sim}{Z} - \underset{\sim}{\beta}) = \text{Var}(\underset{\sim}{Z})$.
   (b) $\text{Cov}(\underset{\sim}{a}^\top \underset{\sim}{Z}, \underset{\sim}{b}^\top \underset{\sim}{W}) = \underset{\sim}{a}^\top \text{Cov}(\underset{\sim}{Z}, \underset{\sim}{W})\underset{\sim}{b}$.
   (c) $\text{Cov}(\underset{\sim 1}{X} + \underset{\sim 2}{X}, \underset{\sim 1}{Z} + \underset{\sim 2}{Z}) = \text{Cov}(\underset{\sim 1}{X}, \underset{\sim 1}{Z}) + \text{Cov}(\underset{\sim 1}{X}, \underset{\sim 2}{Z}) + \text{Cov}(\underset{\sim 2}{X}, \underset{\sim 1}{Z}) + \text{Cov}(\underset{\sim 2}{X}, \underset{\sim 2}{Z})$.
   (d) $\text{Var}(\underset{\sim}{X} \pm \underset{\sim}{Z}) = \text{Var}(\underset{\sim}{X}) + \text{Var}(\underset{\sim}{Z}) \pm \text{Cov}(\underset{\sim}{X}, \underset{\sim}{Z}) \pm \text{Cov}(\underset{\sim}{Z}, \underset{\sim}{X})$.
   (e) $E(\underset{\sim}{Z}^\top A \underset{\sim}{Z}) = \underset{\sim}{\mu}^\top A \underset{\sim}{\mu} + \text{tr}(A\Sigma)$, where $E(\underset{\sim}{Z}) = \underset{\sim}{\mu}$, $\text{Var}(\underset{\sim}{Z}) = \Sigma$.

2. Let $X = ((X_{i,j}))$ be an $m \times n$ matrix. Define $E(X) = ((E(X_{i,j})))$. Let $A = ((a_{i,j}))$ and $B = ((b_{i,j}))$ be non-random matrices. Then show that $E(AXB) = AE(X)B$.

3. Let $X_1, \ldots, X_n$ be independent random variables with $E(X_i) = 0$, $var(X_i) = \mu_2$, $E(X_i^3) = 0$ and $E(X_i^4) = \mu_4$ for all $i = 1, \ldots, n$. Let $A = ((a_{i,j}))$ be a non-random symmetric matrix. Then show that

$$var\left(\underset{\sim}{X}^T A \underset{\sim}{X}\right) = \mu_2^2 \left(2\sum_{i,j} a_{i,j}^2 - 3\sum_i a_{i,i}^2\right) + \mu_4 \left(\sum_i a_{i,i}^2\right).$$

2 Differentiation

1. Find the maxima and/or minima of the following functions:
   (a) $g(\underset{\sim}{x}) = \|\underset{\sim}{y} - A\underset{\sim}{x}\|^2$ where A is a p.d. invertible matrix.
   (b) $g(\underset{\sim}{x}) = \sum_{j=1}^k (x_j^4 - 4x_j^2)$
   (c) $g(\underset{\sim}{x}) = \|\underset{\sim}{x}\|^2 + \exp\{\underset{\sim}{a}^T \underset{\sim}{x}\}$

2. Let (X, Y) have the joint pdf

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

   Define the transformation $U = X + Y,\ V = X - Y$. Find the joint distribution of (U, V).

3. Let (X, Y) have the joint pdf

$$f_{X,Y}(x,y) = \begin{cases} 4xy, & 0 < x < 1,\ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

   Define the transformation $U = XY,\ V = Y$. Find the joint distribution of (U, V).

4. Let (X, Y) have the joint pdf

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)}, & x > 0,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation $U = X, \quad V = Y/X$. Find the joint distribution of (U, V).

5. Let $(X_1, X_2, \ldots, X_k)$ have the joint pdf

$$f_{X_1,\ldots,X_k}(x_1, \ldots, x_k) = \begin{cases} e^{-(x_1+x_2+\cdots+x_k)}, & x_i > 0,\ i = 1, \ldots, k, \\ 0, & \text{otherwise.} \end{cases}$$

Define the transformation

$$U_1 = X_1, \qquad U_i = \frac{X_i}{X_1}, \quad i = 2, \ldots, k.$$

Find the joint distribution of $\{U_1, \ldots, U_n\}$.

3 Multivariate Normal Distribution

1. Let $X \sim \texttt{normal}(0, 1)$. Define $Y = \text{sign}(X)$ and $Z = |X|$. Here $\text{sign}(\cdot)$ is a $\mathbb{R} \to \{0, 1\}$ function such that $\text{sign}(a) = 1$ if $a \geq 0$, and $\text{sign}(a) = -1$ otherwise.
   (a) Find the marginal distributions of Y and Z.
   (b) Find the joint CDF of (Y, Z). Hence or otherwise prove that Y and Z are independently distributed.

2. Let the joint distribution of $\underset{\sim}{X}$ be an equal mixture of $N_4\left(\underset{\sim}{0}, \Sigma_1\right)$ and $N_4\left(\underset{\sim}{0}, \Sigma_2\right)$, where $\Sigma_1 = \begin{bmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,\rho} \end{bmatrix}$ and $\Sigma_2 = \begin{bmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,-\rho} \end{bmatrix}$, and $E_{2,\rho}$ is the 2×2 equi-correlation matrix with $-1 < \rho < 1$.
   (a) Find the joint pdfs of (X_1, X_2) and (X_3, X_4).
   (b) Prove that X_4 and X_3 are uncorrelated but not independent, and
   (c) Are (X_1, X_2) and (X_3, X_4) independent?

3. Let $\underset{\sim}{X} = (X_1, X_2, X_3)^\top$ be a random vector with joint pdf

$$f_{\underset{\sim}{X}}(\underset{\sim}{x}) = (2\pi)^{-3/2} \exp\{-(x_1^2+x_2^2+x_3^2)/2\} \times \left\{1 + x_1 x_2 x_3 \exp\{-(x_1^2 + x_2^2 + x_3^2)/2\}\right\},\ x_i \in \mathbb{R}, \forall i = 1, 2, 3.$$

   (a) Show that $X_i \sim N(0, 1)$ for all $i = 1, 2, 3$.
   (b) Show that (X_i, X_j) jointly follow a bivariate normal distribution. Hence find the expectation and variance-covariance matrix of $\underset{\sim}{X}$.
   (c) Does $\underset{\sim}{X}$ follow a multivariate normal distribution. Justify your answer.

4. Let $\underset{\sim}{Z} = (X, Y)^\top$ be distributed as a bivariate normal distribution with parameters $(1, 2, 2.5, 1.5, 0.5)$, and let $f_{\underset{\sim}{Z}}$ be the pdf of $\underset{\sim}{Z}$. Identify the set $\rho(c) = \{\underset{\sim}{z} : f_{\underset{\sim}{Z}}(\underset{\sim}{z}) \geq c\}$. Draw $\rho(0.05)$ and $\rho(0.025)$.

5. Let $\underset{\sim}{y}$ be a random vector defined as $\underset{\sim}{y} = X\underset{\sim}{\beta} + Z\underset{\sim}{b} + \underset{\sim}{\varepsilon}$, where X and Z are $p \times q$ and $p \times r$ fixed matrices, $\underset{\sim}{\beta}$ is a fixed vector, $\underset{\sim}{b} \sim N_r\left(\underset{\sim}{0}, \Delta\right)$ and $\underset{\sim}{\varepsilon} \sim N_p\left(\underset{\sim}{0}, \sigma^2 I_p\right)$. Further, $\underset{\sim}{b}$ and $\underset{\sim}{\varepsilon}$ are independent. Show that

(a) $\underset{\sim}{y} \sim N_p\left(X\underset{\sim}{\beta}, W\right)$ where $W = Z\Delta Z^T + \sigma^2 I_p$.

(b)

$$\begin{bmatrix} \underset{\sim}{y} \\ \underset{\sim}{b} \end{bmatrix} \sim N_{p+r}\left(\begin{bmatrix} X\underset{\sim}{\beta} \\ \underset{\sim}{0} \end{bmatrix}, \begin{bmatrix} W & Z\Delta \\ \Delta Z^T & \Delta \end{bmatrix}\right).$$

(c) $E\left(\underset{\sim}{b}|\underset{\sim}{y}\right) = \Delta Z W^{-1}\left(\underset{\sim}{y} - X\underset{\sim}{\beta}\right)$.

(d) $\underset{\sim}{y}|\underset{\sim}{b} \sim N_p\left(X\underset{\sim}{\beta} + Z\underset{\sim}{b}, \sigma^2 I_p\right)$.

6. Let $\underset{\sim}{Y} \sim N_p(\underset{\sim}{\mu}, \Sigma)$ where $\Sigma > 0$. Define the moment generating function (MGF) of $\underset{\sim}{Y}$ at $\underset{\sim}{t}$ as $M_{\underset{\sim}{Y}}(\underset{\sim}{t}) = E\left(\exp\{\underset{\sim}{t}^\top \underset{\sim}{Y}\}\right)$. Without explicitly deriving, find $M_{\underset{\sim}{Y}}(\underset{\sim}{t})$ using the properties of normal distribution and MGF of univariate normal distribution.

7. Let $\underset{\sim}{Y} \sim N_p(\underset{\sim}{\mu}, I)$, and U be a rectangular matrix with $UU^\top = I_q$, $q < p$. Find the distribution $U\underset{\sim}{Y}$.

8. Suppose that $Y_1, Y_2, \ldots, Y_n$ are independently distributed as $N(0,1)$. Calculate the characteristic function of the random vector

$$\left(\overline{Y},\, Y_1 - \overline{Y},\, Y_2 - \overline{Y},\, \ldots,\, Y_n - \overline{Y}\right)$$

and hence show that $\overline{Y}$ is independent of $\sum_{i=1}^n (Y_i - \overline{Y})^2$.

9. Let X_1, X_2, and X_3 be i.i.d. $N(0,1)$. Let

$$Y_1 = \frac{X_1 + X_2 + X_3}{\sqrt{3}}, \quad Y_2 = \frac{X_1 - X_2}{\sqrt{2}}, \quad Y_3 = \frac{X_1 + X_2 - 2X_3}{\sqrt{6}}.$$

Show that Y_1, Y_2 and Y_3 are i.i.d. $N(0,1)$.

(The transformation above is a special case of the so-called *Helmert transformation.*)

10. Given $\underset{\sim}{Y} \sim N_3(\mu, \Sigma)$, where

$$\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix},$$

for what value(s) of ρ are $Y_1 + Y_2 + Y_3$ and $Y_1 - Y_2 - Y_3$ statistically independent?

11. Let $\underset{\sim}{Y} \sim N_n(\underset{\sim}{\theta}, \sigma^2 I)$ and let $Q_i = (\underset{\sim}{Y} - \underset{\sim}{\theta})' P_i (\underset{\sim}{Y} - \underset{\sim}{\theta})/\sigma^2$, $i = 1, 2$.

Show that if $Q_i \sim \chi^2_{r_i}$ and $Q_1 - Q_2 \geq 0$, then $Q_1 - Q_2$ and Q_2 are independently distributed as $\chi^2_{r_1 - r_2}$ and $\chi^2_{r_2}$, respectively.

12. Suppose that $\underset{\sim}{Y} \sim N_3(0, I)$. Show that

$$\frac{1}{3}\left[(Y_1 - Y_2)^2 + (Y_2 - Y_3)^2 + (Y_3 - Y_1)^2\right]$$

has a χ^2_2 distribution. Does some multiple of

$$(Y_1 - Y_2)^2 + (Y_2 - Y_3)^2 + \cdots + (Y_{n-1} - Y_n)^2 + (Y_n - Y_1)^2$$

have a chi-squared distribution for general n?